One process, two threads

Which parts of one job can run at the same time?

Names: _____ Partner: _____

20 minutes: setup 2 · pairs 12 · comparison 5 · exit 1. Spend about 4 minutes on each of A, B and C. Explain locations and ownership; do not calculate counts or execution times.

The sensor program

Two threads run the same count function on different chunks of a readings buffer. Each call has local variables `i` (reading position) and `count` (running tally). The threshold is a global variable. The buffer was allocated in the heap when the program ran.

Process P: one address space

Code count function	**Globals** threshold	**Heap** readings buffer	**Stack A** i, count	**Stack B** i, count

Thread A execution state PC and registers	**Thread B execution state** PC and registers

PC means program counter: next-instruction address. The state boxes are not extra memory regions. The region widths are schematic, not memory sizes.

A. Locate and classify (4 minutes)

Write **shared by the threads** or **own to one thread**. Name the region or state record that supports each answer.

Item	Shared or own?	Location / reason
Threshold global		
Readings buffer		
Local i in A's count call		
Count function's code		
A's PC		

In these cases, no thread deliberately accesses another thread's stack. “Own” describes separate execution state/local storage; it does not claim OS memory protection between threads.

Reason from the memory picture

> Consider each change separately. Pause counting, complete the change, then let the other thread read; assume the completed change is visible. No simultaneous updates or special sharing between processes are involved.

B. Does the other execution see the change? (4 minutes)

1. A changes P's threshold global. B then reads P's threshold. Does B read the changed threshold? Why?

2. A advances local i in its own count call. Does B's local i advance too? Why?

3. A in process P changes P's threshold. Another thread runs the same program in a separate process Q. Does Q's threshold change too? Sketch Q's separate address space as part of your reason.

C. Can these steps run together? (4 minutes)

The file has been read. Input stays read-only; each thread writes only its own partial result. Two compute cores are available.

1. Count chunk A and count chunk B: may they execute at the same time? Explain inputs, output locations and cores.

2. Combine the two results and count an unfinished chunk: may Combine start now? Explain which result is missing.

Comparison (5 minutes): Point to the diagram for each reason. Compare the one-process case with the two-process sketch.

Exit (1 minute)

Define **process** and **thread**, one sentence each.